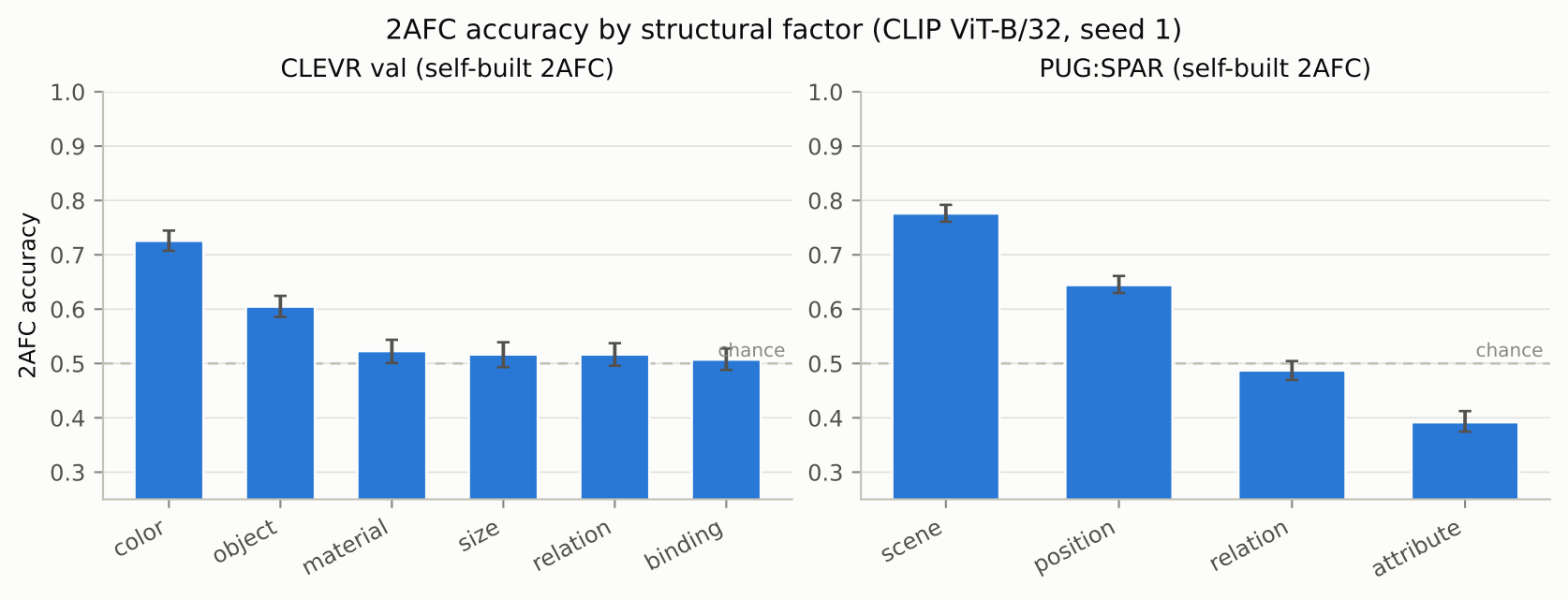
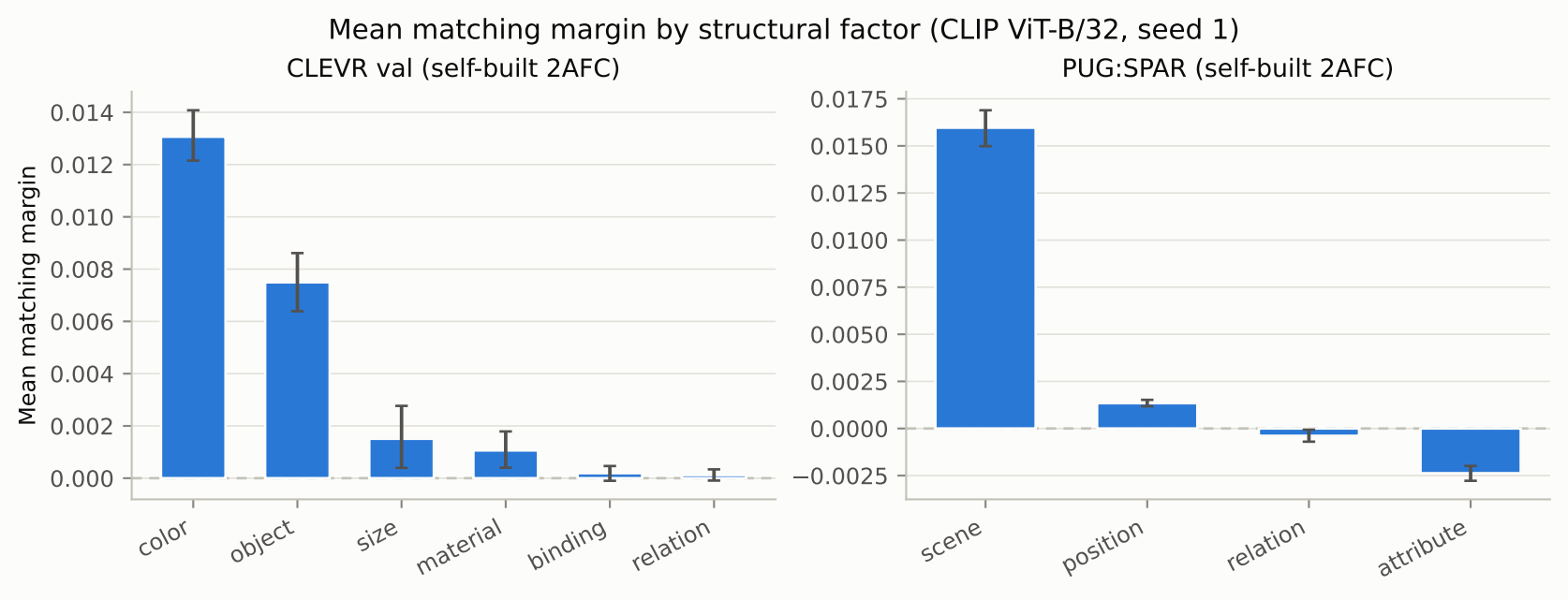
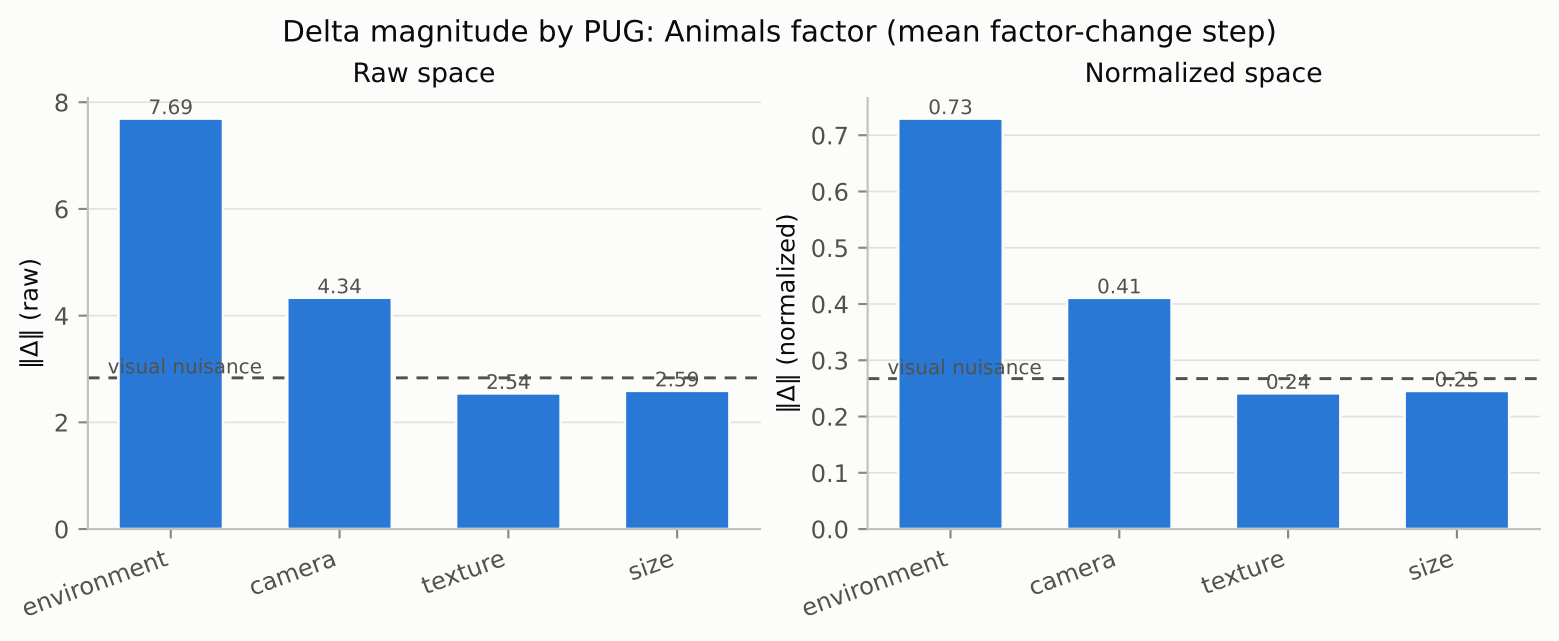
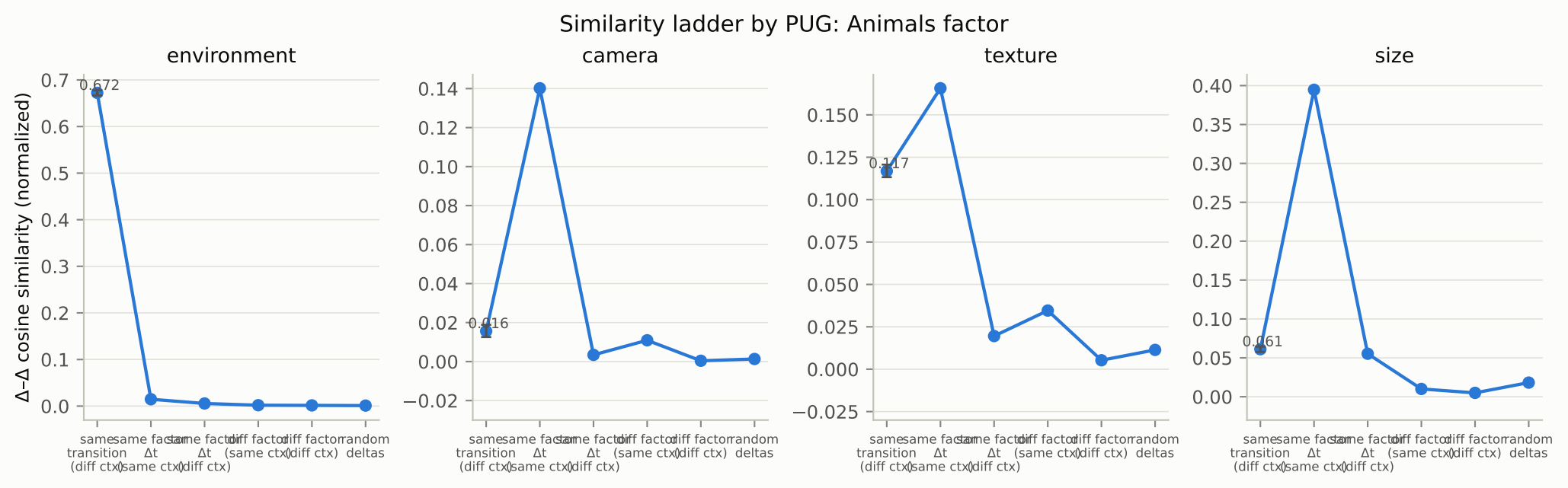
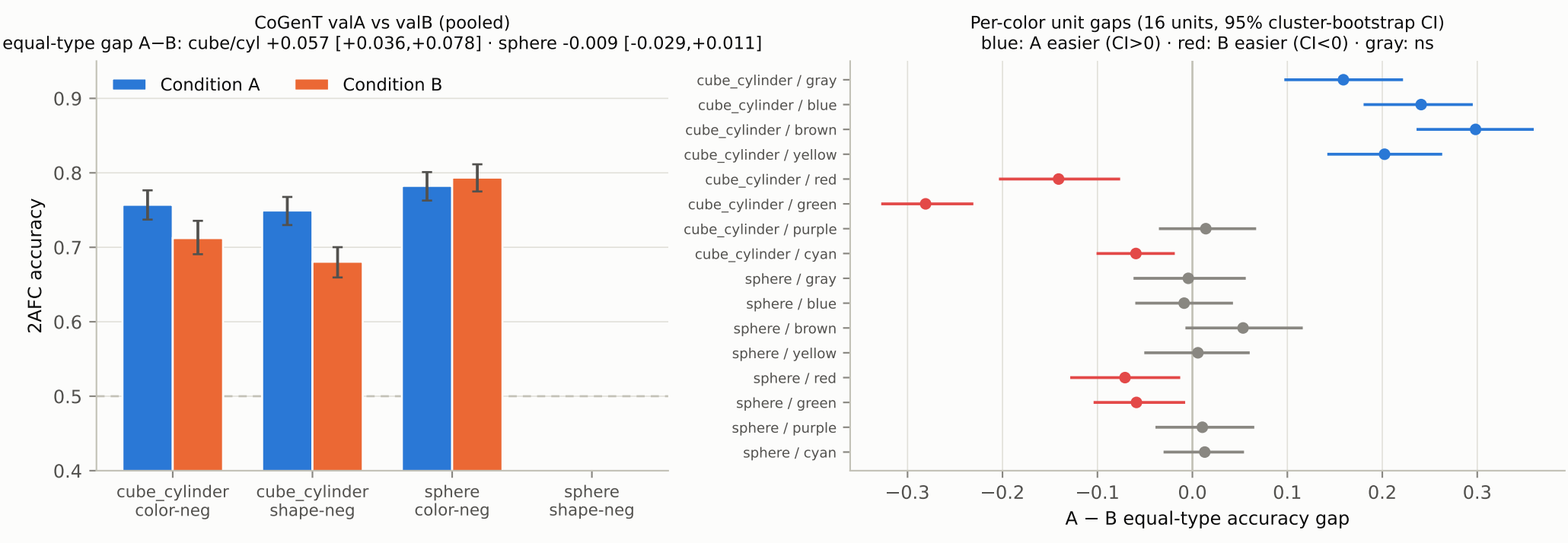

2AFC accuracy by structural factor (CLIP ViT-B/32, seed 1)
CLEVR val (self-built 2AFC)
PUG:SPAR (self-built 2AFC)
2AFC accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
chance
color
object
material
size
relation
binding
scene
position
relation
attribute


Mean matching margin by structural factor (CLIP ViT-B/32, seed 1)
CLEVR val (self-built 2AFC)
PUG:SPAR (self-built 2AFC)
Mean matching margin
0.014
0.012
0.010
0.008
0.006
0.004
0.002
0.000
color
object
size
material
binding
relation
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
−0.0025
scene
position
relation
attribute


Delta magnitude by PUG: Animals factor (mean factor-change step)
Raw space
‖Δ‖ (raw)
8
6
4
2
0
7.69
4.34
2.54
2.59
visual nuisance
environment
camera
texture
size
Normalized space
‖Δ‖ (normalized)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.73
0.41
0.24
0.25
visual nuisance
environment
camera
texture
size


Similarity ladder by PUG: Animals factor
environment
camera
texture
size
Δ–Δ cosine similarity (normalized)
0.672
0.016
0.117
0.061
same transition (diff ctx)
same factor Δt (same ctx)
same factor Δt (diff ctx)
diff factor (same ctx)
diff factor (diff ctx)
random deltas


CoGenT valA vs valB (pooled)
equal-type gap A−B: cube/cyl +0.057 [+0.036,+0.078] · sphere -0.009 [-0.029,+0.011]
Condition A
Condition B
2AFC accuracy
0.9
0.8
0.7
0.6
0.5
0.4
cube_cylinder color-neg
cube_cylinder shape-neg
sphere color-neg
sphere shape-neg
Per-color unit gaps (16 units, 95% cluster-bootstrap CI)
blue: A easier (CI>0) · red: B easier (CI<0) · gray: ns
cube_cylinder / gray
cube_cylinder / blue
cube_cylinder / brown
cube_cylinder / yellow
cube_cylinder / red
cube_cylinder / green
cube_cylinder / purple
cube_cylinder / cyan
sphere / gray
sphere / blue
sphere / brown
sphere / yellow
sphere / red
sphere / green
sphere / purple
sphere / cyan
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
A − B equal-type accuracy gap